Phase 1 Review Report — CortexHR AI/ML Employee Attrition

Undergraduate Technical Review | B.Tech Computer Science | CortexHR Project

Student Name:	Dikshit Kumar	**Track / Phase:**	AI/ML | Phase 1
Project:	CortexHR AI/ML — Employee Attrition	**Git Branch:**	master (Tag: pending)
Repository:	https://github.com/dikshit0012/CortexHR-.git	**Evaluation Integrity:**	Labels strictly sealed during training

1. Summary of Phase 1 Tasks

Phase 1 encompassed five interconnected milestones establishing a leak-free data audit, an automated feature engineering pipeline, rigorous baseline classification models, calibrated risk banding, and unsupervised attendance anomaly detection.

Task	Status	What I Worked On	Verified Evidence
Task 1: Data Audit & Exploration	**Complete & Tested**	Explored raw dataset (2,389 rows, 42 columns). Detected 6 duplicate rows, 4 negative tenure errors, and 6 severe target leakage columns (resignation date, exit reason, etc.). Assessed class imbalance (14.7% attrition rate).	Task 1/cortexhr-ml/reports/findings.json; Task1_Data_Audit_Report.pdf
Task 2: Reusable Feature Pipeline	**Complete & Tested**	Engineered clean scikit-learn ColumnTransformer pipeline. Dropped duplicates and invalid tenure rows to yield 2,379 clean employee records. Joined 3-month rolling attendance/payroll aggregates strictly inside cross-validation folds.	task 2/files/extracted/cortexhr-ml-task5/reports/tables/day2_pipeline_run.json; src/cortexhr/features.py
Task 3: Train Honest Baselines	**Complete & Tested**	Trained baseline models using 5-fold stratified CV: Dummy (always stays), balanced Logistic Regression, and Decision Trees. Validated accuracy paradox under class imbalance and diagnosed severe decision tree overfitting.	task 4/files/cortexhr-ml/reports/tables/day3_baseline_comparison.csv; reports/day3_baselines.md; ROC-AUC: 0.712 vs 0.713 benchmark (2,379 real rows — not the 2,400-row synthetic stand-in)
Task 4: Evaluation & Risk Bands	**Complete & Tested**	Evaluated out-of-fold predictions. Calibrated probabilities with sigmoid scaling (Brier score 0.206 -> 0.116). Calculated top-decile lift (2.26x) and defined 3 actionable operational risk bands (High, Medium, Low). Authored executive summary.	task 4/files/day4_metrics_table.csv; task 4/files/Day4_Manager_Note.pdf; 29 unit tests passing
Task 5: Unsupervised Attendance Anomaly Detector	**Complete & Tested**	Trained Isolation Forest (3% contamination) on 7,339 employee-months without labels. Combined with domain rules (unexcused absence, impossible hours). Measured against hidden labels using catch_rate(). Built HR priority tiers.	task 5/reports/task5_metrics.csv; task 5/reports/task5_missed_anomalies.csv; 14/14 unit tests passing

2. Proof That the Work Runs & Verified Metrics

All code, test suites, and evaluation notebooks were executed against the real CortexHR dataset. No benchmark metrics were copied as my own; every figure reported below reflects verified local execution.

A. Data Audit & Feature Pipeline (Tasks 1 & 2)

• **Dataset Intake:** Audited 2,389 raw rows across 42 columns. Dropped 6 duplicate employee records (6 dropped (EMP-0060, EMP-0237, EMP-0238, EMP-0325, EMP-0483, EMP-0570)) and excluded 4 impossible negative tenure entries (4 dropped (EMP-0248, EMP-0280, EMP-0568, EMP-0596)).
• **Target Leakage Protection:** Identified and dropped 6 exit-related columns (6 columns excluded (resignation_date, exit_date, exit_reason, etc.)).
• **Pipeline Clean Output:** Yielded 2,379 clean records (349 leavers, 14.67% base prevalence). Engineered rolling 3-month attendance aggregates joined strictly inside CV folds to prevent temporal data leakage.

B. Honest Baseline Classification (Task 3 — 5-Fold Stratified CV)

Model	ROC-AUC (Ours)	PR-AUC	Precision	Recall	Accuracy	Reference Benchmark
Dummy (always stays)	0.500 +/- 0.000	0.147	0.000	0.000	0.853	0.854 benchmark
Logistic Regression (balanced)	0.712 +/- 0.018	0.309	0.265	0.633	0.686	0.713 benchmark
Decision Tree (depth 4, balanced)	0.654 +/- 0.038	0.255	0.236	0.525	0.681	0.650 benchmark
Decision Tree (unconstrained)	0.542 +/- 0.028	0.164	0.214	0.226	0.765	Overfit (train ROC: 0.99+)

C. Probability Calibration & Operational Risk Bands (Task 4)

• **Calibration Impact:** Raw balanced Logistic Regression outputs ranking scores rather than true probabilities. Applying sigmoid calibration dropped the Brier score from **0.206** to **0.116**, substantially outperforming the naive base rate baseline (0.125).
• **Decile Lift:** The top 10% scored segment (238 employees) attained a **2.26x lift** (33.2% leaver concentration vs 14.7% baseline) capturing 22.6% of all eventual leavers.
• **Unit Tests:** All 29 evaluation test cases in test_evaluation.py executed cleanly.

D. Unsupervised Attendance Anomaly Detection (Task 5)

Detector Approach	Precision	Recall (Catch Rate)	F1 Score	TP	FP	FN	Flagged Records
Published Baseline (anomaly_baselines)	0.403	0.593	0.480	89	132	61	221 (3.01%)
Baseline Recipe Re-run (raw 10 cols, 3%)	0.398	0.587	0.474	88	133	62	221 (3.01%)
Isolation Forest (curated features)	0.412	0.607	0.491	91	130	59	221 (3.01%)
Rule-Based Only (domain rules)	0.918	0.600	0.726	90	8	60	98 (1.34%)
Combined (IF OR Business Rules)	0.488	0.827	0.614	124	130	26	254 (3.46%)

• **Evaluation Protocol:** Predictions were generated and written to disk without labels. Ground-truth labels were only loaded afterward inside evaluation.py.
• **Anomaly Types Caught:** 100% recall on ABSENCE_STREAK (40/40), IMPOSSIBLE_HOURS (35/35), SUDDEN_DROP (30/30), and ZERO_ATTENDANCE (15/15).
• **Unit Test Verification:** 14 of 14 unit tests passed in tests/test_attendance_anomaly_detector.py.

3. What I Learned (Student Reflection)

"During Phase 1, I learned that practical machine learning is mostly about data discipline rather than complex algorithms. I saw firsthand how target leakage columns like resignation dates can produce deceptively perfect results if left in the dataset, and why an 85% accuracy baseline is completely misleading when only 14.7% of employees leave. Calibrating probabilities taught me that raw classification scores cannot be handed to business managers without adjustment, and building the attendance detector showed me how unsupervised isolation trees and simple business rules can catch anomalies effectively without ever training on ground-truth labels."

4. Real Engineering Challenges & Solutions

1. Subtle Target Leakage in Raw Employee Features

Problem: During Task 1 exploration, several columns like 'resignation_date', 'exit_date', and 'exit_reason' had over 85% missing values. Initially, it looked like standard missing data. However, checking the target label revealed that these columns were populated ONLY for employees who left the company.

Investigation & Resolution: Keeping these columns would allow a model to achieve 100% training accuracy through pure hindsight leakage. I classified all 6 exit-related columns as 'LEAK' in the data dictionary and strictly excluded them from the feature pipeline.

2. High Training Score vs Collapsing Cross-Validation in Decision Trees

Problem: When running baseline models in Task 3, an unconstrained Decision Tree appeared to learn the training split almost perfectly (training ROC-AUC > 0.99). However, its 5-fold cross-validation ROC-AUC collapsed to 0.542, performing barely better than random coin-flipping.

Investigation & Resolution: I ran a depth-tuning experiment across max_depth values 1 through 8. Constraining max_depth to 4 prevented the tree from memorizing small sample idiosyncrasies, yielding a stable cross-validation ROC-AUC of 0.654 +/- 0.038 and matching the benchmark.

3. Single-Feature Anomaly Invisibility in Multi-Dimensional Isolation Forest

Problem: In Task 5, our unsupervised Isolation Forest caught almost all absence streak, sudden drop, and impossible hours anomalies, but missed 26 out of 30 LATE_SPIKE records (13.3% recall). Examining the missed records revealed that employees with late arrival spikes had completely normal attendance rates (0.98) and standard working hours (8.55 hrs/day).

Investigation & Resolution: Because 8 out of 9 features were completely normal, random isolation partitioning did not isolate these records early enough to beat the 3% cut-off. I documented this as an inherent limitation of tree-based multi-feature isolation for one-dimensional spikes, and designed a priority ranking system that combines rule triggers with model scores to flag such cases for HR

review.

5. Roadmap for Phase 2 (Payroll Anomaly Detection)

In Phase 2, detection moves from attendance to payroll, where data errors translate directly into financial discrepancy and employee dissatisfaction. Building upon Phase 1 foundation:

• 1. Reusing Established Pipeline Components: Port the clean ColumnTransformer structure and rolling window join logic from Task 2 into the Phase 2 payroll workflows to ensure zero data leakage.

• 2. Payroll Feature Engineering: Construct monthly payroll indicators (such as overtime-to-base salary ratios, retroactive adjustments, and sudden compensation rate shifts) that capture financial anomalies.

• 3. Dual-Layer Anomaly Detection on Payroll: Apply Isolation Forest paired with strict statutory/labour rules (such as minimum wage violations, negative net pay, or excessive monthly overtime hours).

• 4. Rigorous Out-of-Fold Evaluation: Keep validation labels strictly sealed in the vault until predictions are finalized, evaluating detector performance via precision, recall, and financial catch rate.

• 5. Actionable Payroll Review Tiers: Implement a points-based prioritization queue so payroll accountants can audit High-risk payroll batches first, minimizing financial error risk before disbursement.